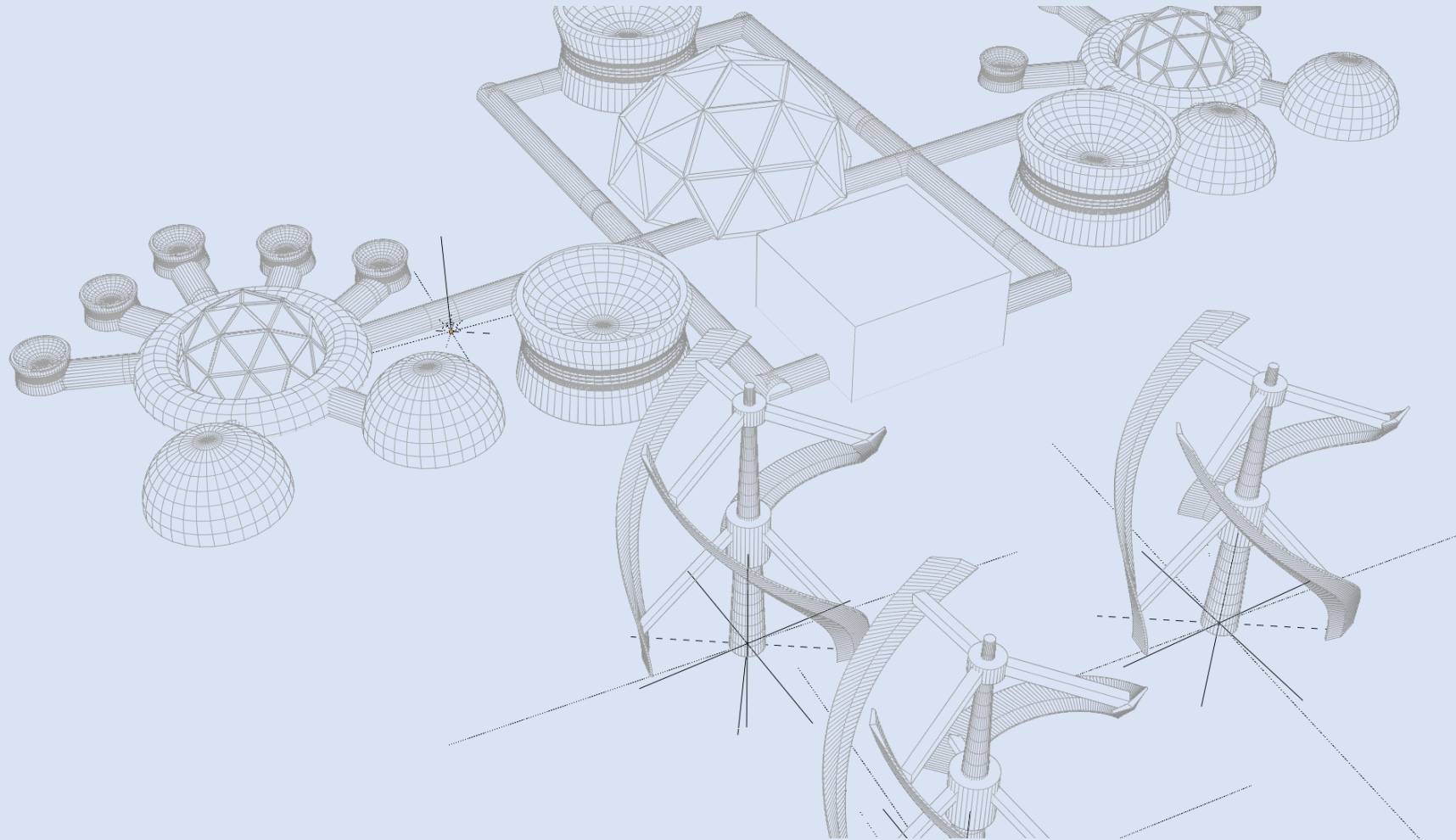


TEAM CELESTIAL



Team members:

Muhir Kapoor

Mission Concept & Roadmap

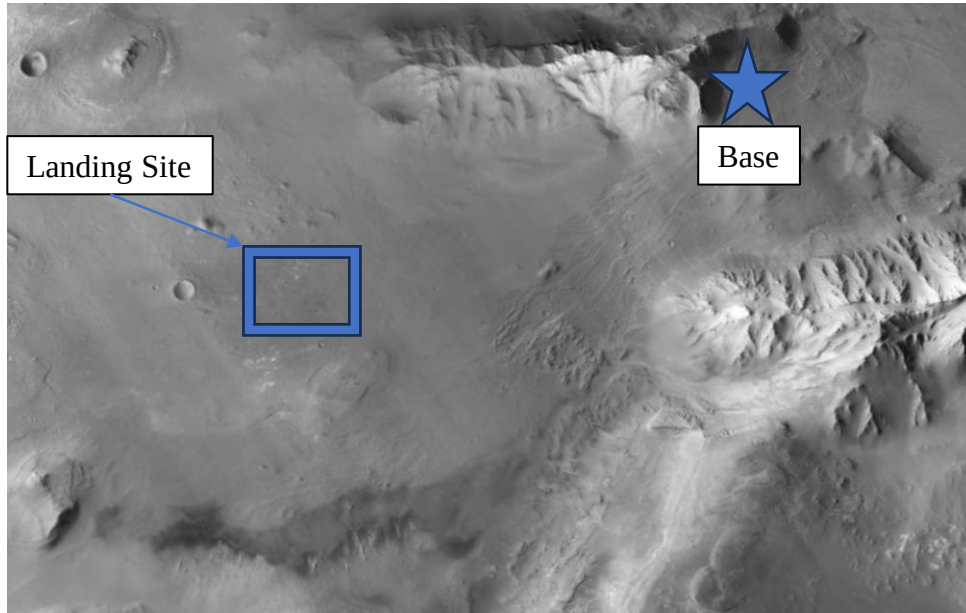


Figure 1.1: Landing Site based in close proximity to Base Location
Source: Murray lab, Caltech

Heavy lift rockets utilizing nuclear thermal propulsion, Hypersonic Inflatable Aerodynamic Decelerators and Precision landing technologies would transfer materials, equipment and crew to mars sequentially.

A swarm of rovers would be deployed to set up the communication systems and power generation, transmission and distribution system.

The rovers would then mine and transport minerals and ores to industrial refineries to produce water, oxygen, construction and other important materials for the base via technologies discussed in the aforementioned sections.

BASE LOCATION SELECTION AND JUSTIFICATION

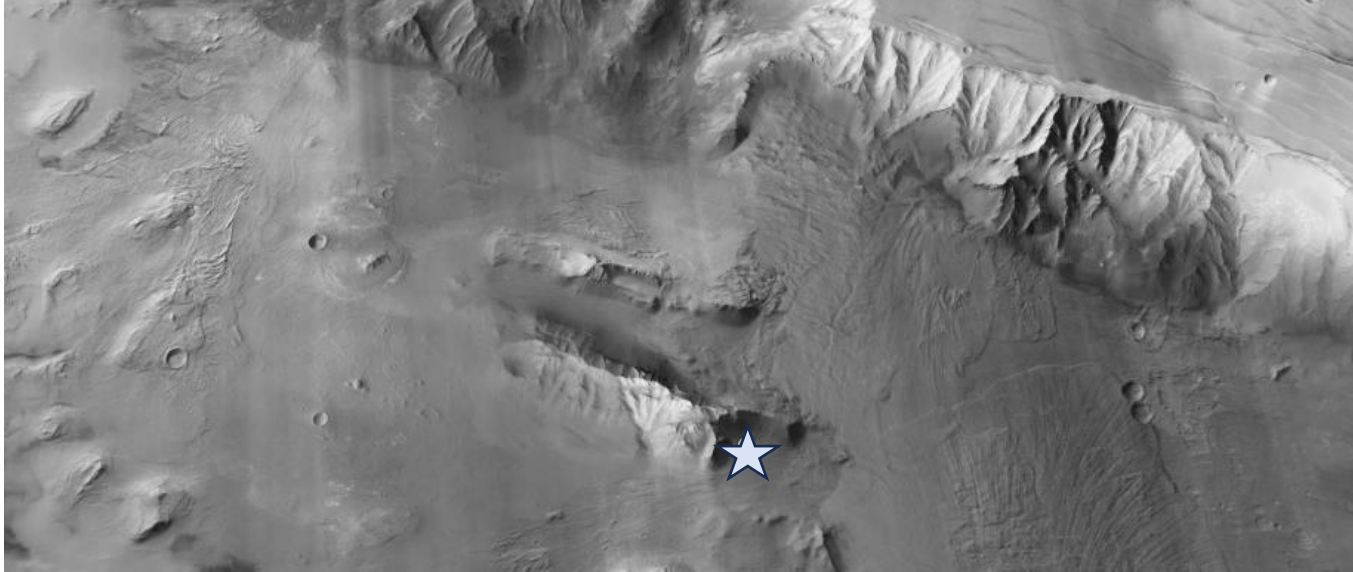


Figure 1.2: Location of the base on Eastern Melas Chasma (Star)

Source: Murray lab, Caltech

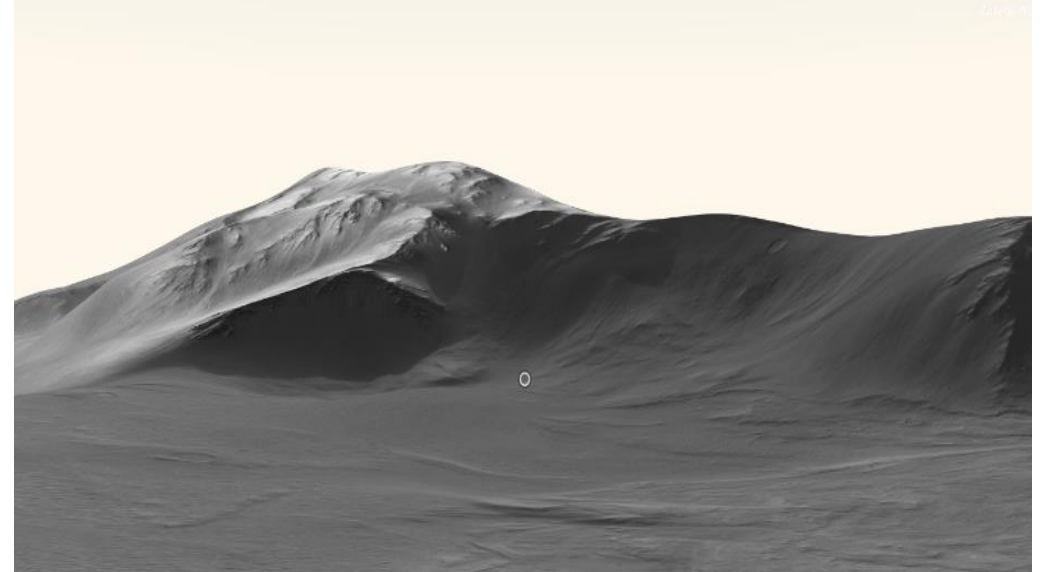


Figure 1.3: Closeup of the location on Eastern Melas Chasma,

Source: Murray lab, Caltech

Eastern Melas Chasma, Valles Marineris

- Coordinates: 11.6°S, 68.562°W
- Elevation: **-4.6 km, 1.2 kPa**
- Solar Flux: **150 W /m²**
- Temperature range: **-70°C-30°C**
- Wind Speed: 35 m/s (near slopes)

-Increased atmospheric pressure (~1200 Pa), facilitates easy entry, descent, and landing of large payloads.

-Massive canyon walls and flat-topped mesas coupled with higher atmospheric pressure in eastern Melas Chasma offer significant protection against:

- Radiation.
- Micrometeorite impacts.

Figure 1.4: Resource rubric
Source: Eastern Melas Chasma EZ; [2] NASA

BASE LOCATION SELECTION AND JUSTIFICATION

Site Factors				RROI-1	EZ SUM
ISRU and Civil Engineering Criteria	Engineering	Meets First Order Criteria (Latitude, Elevation, Thermal Inertia)		●	●
	Water Resource	Threshold	Potential for ice or ice/regolith mix	●	●
			Potential for hydrated minerals	●	●
			Quantity for substantial production	●	●
			Potential to be minable by highly automated systems	●	●
			Located less than 3 km from processing equipment site	●	●
			Located no more than 3 meters below the surface	●	●
			Accessible by automated systems	●	●
		Qualifying	Potential for multiple sources of ice, ice/regolith mix and hydrated minerals	●	●
			Distance to resource location can be >5 km	●	●
			Route to resource location must be (plausibly) traversable	●	●
	Civil Engineering	Threshold	~50 sq km region of flat and stable terrain with sparse rock distribution	●	●
			1–10 km length scale: <10°	●	●
			Located within 5 km of landing site location	○	○
		Qualifying	Located in the northern hemisphere	●	●
			Evidence of abundant cobble sized or smaller rocks and bulk, loose regolith	●	●
	Food Production	Qualifying	Utilitarian terrain features	○	○
			Low latitude	●	●
			No local terrain feature(s) that could shadow light collection facilities	●	●
			Access to water	●	●
	Metal/Silicon Resource	Threshold	Access to dark, minimally altered basaltic sands	●	●
			Potential for metal/silicon	●	●
			Potential to be minable by highly automated systems	●	●
			Located less than 3 km from processing equipment site	●	●
			Located no more than 3 meters below the surface	●	●
			Accessible by automated systems	●	●
		Qualifying	Potential for multiple sources of metals/silicon	●	●
			Distance to resource location can be >5 km	●	●
			Route to resource location must be (plausibly) traversable	●	●

Key	
●	Yes
○	Partial Support or Debated
	No
?	Indeterminate

Resources

Water

- Polyhydrated Sulphates hold **$10^{12} m^3$ of water** in forms of hexahydrite and kieserite
- Water fogs in valles marineris
- Possibly RSLs

Metals

- Mafic Bedrock
 - Martian Regolith
 - Basaltic and Aeolian sands
- Rich in ***Fe, Al, Si, Ti, Mg***

BASE PURPOSE AND ACTIVITIES

Aerial Swarm Robots (MarsBees)

(Source: NASA)

- Terrain mapping
- Biosignature identification
- Atmospheric & Climatic data collection



Aerial Drones (Mars Choppers)

(Source: JPL, NASA)

- Transportation of heavy payloads
- Long Distance Research & Exploration
- Support for ground missions



Robotic Architecture

Terrestrial Swarm Robots

(Source: Hassell Studio)

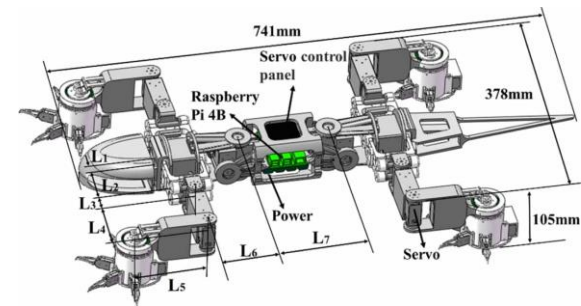
- Resource Identification
- Resource Extraction
- Infrastructure development



Terrestrial lizard-based rovers

(Source:[6])

- Climbing steep terrain
- Navigating through sands
- Resource identification



BASE PURPOSE AND ACTIVITIES

Activities

Research & Exploration

- Subsurface Exploration
- Microbial & Organic Search
- Meteorological Studies
- Human & Dust Centric Insights

Resource Extraction & Manufacturing

- Water Extraction
- Oxygen Production
- Hydrocarbon Production
- Metal extraction, refining and manufacturing

Further space exploration

- In-situ propellant production
- Establishment of staging area (Launchpads & Landing-pads)
- Test-bed for advanced technologies

BASE PURPOSE AND ACTIVITIES

Subsurface Exploration

Goals: Identification of ice deposits in Valles Marineris (e.g., central Candor Chasma), minerals, **lava tubes** & analyze soil stratification

Activities: Usage of **ground-penetrating radar** and swarm robotics to explore lava tubes, ice deposits, and mineral resources.

Contribution: Lava tubes offer potential for future bases, and mapping water and minerals enhances ISRU.

Meteorological and Seismic Substation

Goals: Study dust behavior, surface thermal dynamics, wind patterns, Mars quakes, and geothermal energy potential.

Activities: Establish **meteorological monitoring and geological center**.

Contribution: Data informs dust-resistant equipment, renewable energy systems, and energy-efficient habitats.

Goal 1) Research & Exploration

Microbial and Organic Search

Goals: Investigation of Martian microbes near RSLs in **eastern melas chasma**. and testing microbial survival in Martian conditions,

Activities: Sample subsurface layers for biosignatures and monitor microbial survival.

Contribution: Detecting life informs **planetary protection** and boosts exploration interest. Microbial studies support life-support, waste recycling, and ore extraction systems.

Human and Dust-Centric Insights

Goals: Address challenges of dust infiltration, machinery durability, human mobility, and effects of Martian gravity.

Activities: Monitor dust impacts on systems, test movement on loose soil, and study long-term effects of Martian gravity on human physiology.

Contribution: Improves equipment reliability and human adaptability, ensuring mission success and enabling larger settlements

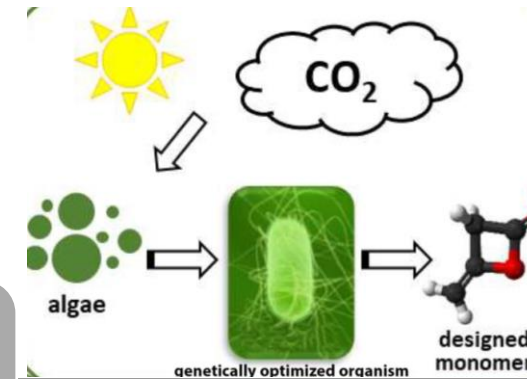
BASE PURPOSE AND ACTIVITIES

Water Extraction [7] [8]

- Hydrated Minerals: Microwave extraction
- Water Fogs: Water Vapor Adsorption Reactor will be used to adsorb water from water fogs.
- RSL's: open air heating.

Hydrocarbon Production [10]

Gene edited algae and bacteria to produce hydrocarbons **with minimal consumption of water resources**

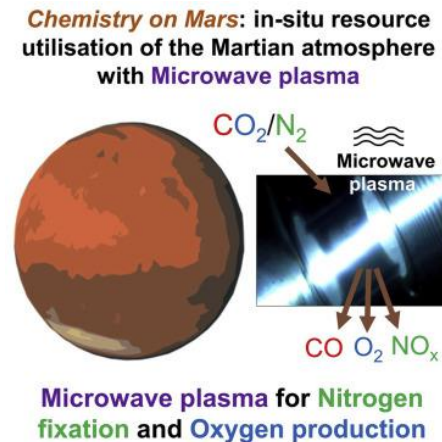


- Production of
- propellants,
 - bioplastics,
 - biopolymers
 - biofuels.

Goal 2) Research & Exploration

Oxygen Production [9]

- Plasma based O₂ , NO₂ and CO generation(30 times quicker than currently achieved with MOXIE)
- O₂ used for life support
- NO₂ used as a fertilizer
- CO used as a propellant



Basaltic Sands, Metal Extraction, Refining, Regolith [11]

Excavation and drilling via swarm robots (if required) proceeded by (if required) **Biomining** &/ thermal reduction &/ smelting &/ electrolysis &/ compression &/ sintering

Extracted metals, minerals, and processed regolith are used for producing solar cells, windmills, **Na-ion batteries**, roads, launchpads, construction elements, fertilizers, medicines, vehicle components, drones, pipelines, electronics, and power transmission lines. **Basalt fibers** reinforce structures, and **cast basalt** is used for construction.

BASE PURPOSE AND ACTIVITIES

Goal 3) Further Space Exploration

In-situ propellant production

- Liquid CO, H₂, O₂

Establishment of staging area (Launchpads & Landing-pads) & **mass driver (2060 onwards)**.

- Launchpads and debris shields constructed via sintering and metals. Rocket engines are 3D printed. **Mass driver is constructed**

Test-bed for advanced technologies

- Offers a unique environment for testing advanced technologies.

- **Mars' proximity to asteroids and outer planets**
 - **Its low gravity**
 - **Its low atmospheric pressure**
- Makes launching easier and reduces mission time.**

FIRST HABITAT FOR 4-8 PEOPLE

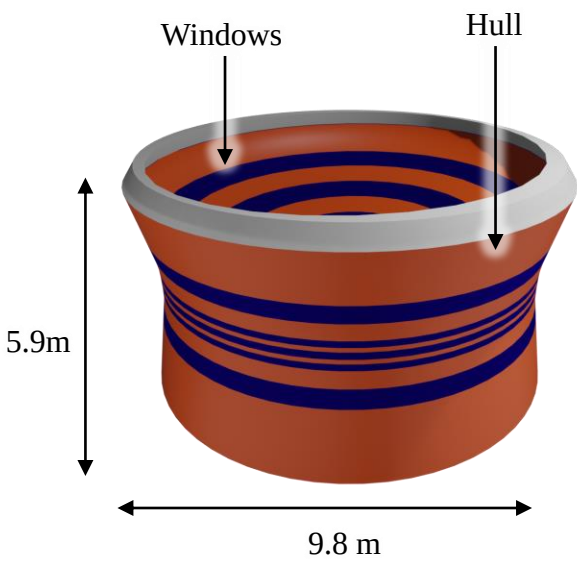


Figure 2.1:
Structure
Source: Muhir Kapoor, Blender

Material	Source	Properties	Usage
Compressed Martian Regolith	Martian Regolith	Compressive strengths that of reinforced concrete (~50MPa) Radiation and Meteorite Shielding Thermal Insulation	Casing of structures, pipes, storage vessels,
Basalt Fibres	Dark Basalt Sands in <i>Eastern Melas Chasma</i>	High Tensile Strength (4800 Mpa) Temperature Resistant	Tensile Reinfortment for compressed martian regolith in pressure vessels
Cast basalt		Non-porous Fire, chemical, moisture resistant	Tiling for floors and walls, inner vessel for pipes, storage vessels
Fused Silica	Interior Layered Deposits, <i>Eastern Melas Chasma</i>	High thermal stability, resistant to thermal shock Resistant to scratching, chemically inert	Windows in greenhouses, structures
Transparent Silica Aerogel	Earth	Superior thermal insulation	Sandwiched in windows, Traps heat

Table 2.1: Materials, Sources, Properties

- Inspired from Model 1 from the paper *Martian buildings: structural forms using in-place sources*.
- Minimizes the tensile under loads such as gravity, wind, marsquakes, asteroid and meteoroid impact loads**
- Natural lighting will be utilized wherever possible.

FIRST HABITAT FOR 4-8 PEOPLE

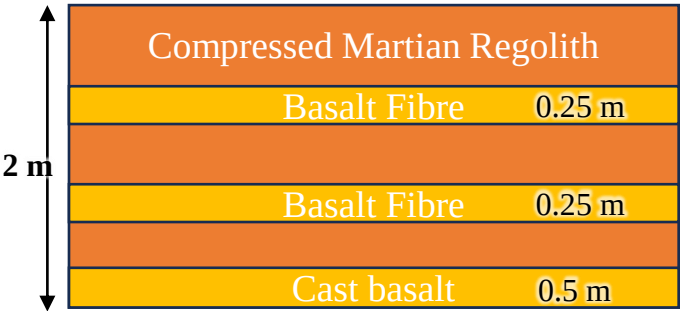


Figure 2.2: configuration of hull
Source: Muhir Kapoor

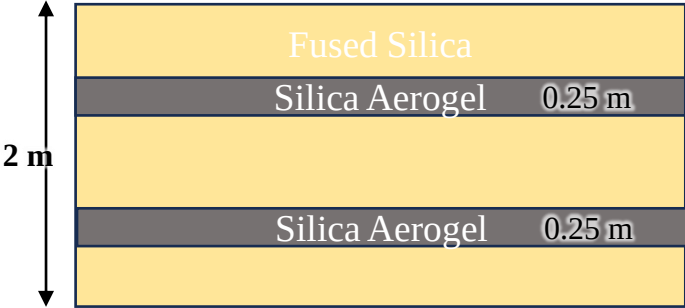


Figure 2.3: configuration of window
Source: Muhir Kapoor

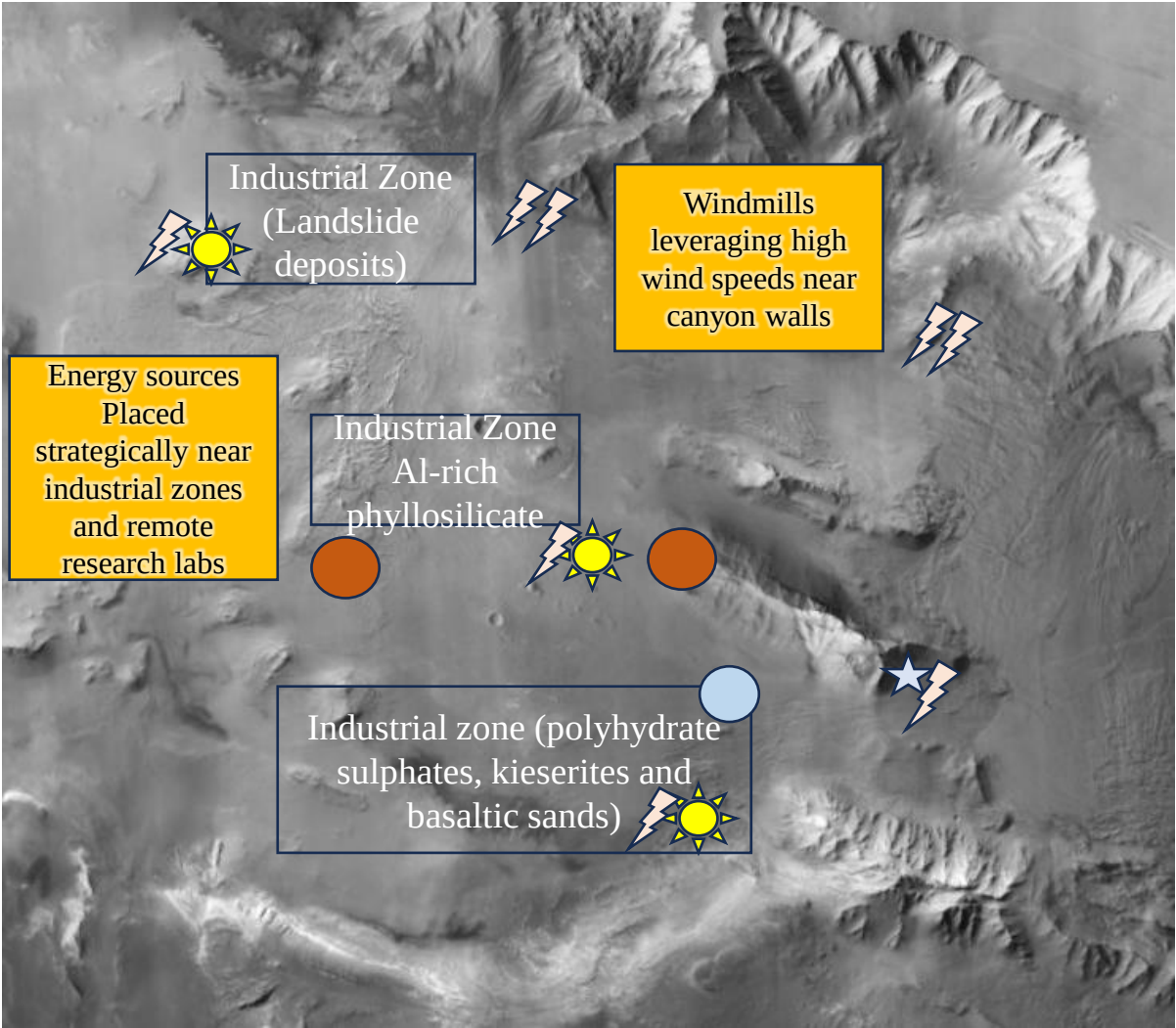


Figure 2.4: Layout of Base
Source: Murray Lab, Caltech

Symbol	Key
	Solar energy Farm
	Wind energy farm
	Residential zone
	Landing sites
	Research Zones

FIRST HABITAT FOR 4-8 PEOPLE

Primary Power Source:

Photovoltaic thermal hybrid solar collectors

- Utilizes active sun trackers
- Produces thermal energy alongside electrical energy, utilized for heating in residential and industrial areas
- Lotus dust mitigation/electrodynamic shield hybrid coating utilized for dust mitigation

Secondary Power Source:

Triboelectric/Turbine hybrid Wind generators [13]

- In dust storms when speeds in Valles Marineris reach up to (~50m/s) ; acts as an alternative source to solar energy.

Compressed energy storage for long term purposes, especially dust storms.

Solid state batteries will be used for overnight storage.

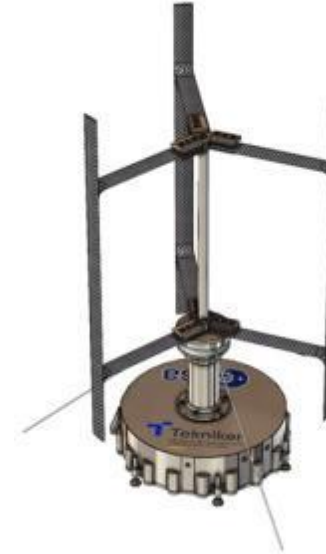


Figure 2.5:Triboelectric Wind Generator
Source: ESA

Laser optical based communication systems offering high data transfer rates, along with **radio wave-based communications** will be used to communicate with earth.

A relay communication satellite will be placed at a **non Keplerian orbit** ensuring that communication between mars and earth remains alive.

Area	Allocation
Residential	200 kW
Operational	90 kW
Industrial	100 kW
Total	390 kW

Table 2.2: Power Requirements

FIRST HABITAT FOR 4-8 PEOPLE

Pressurization of habitat at 0.75 atm to reduce stress on walls.

Component	N ₂	O ₂	H ₂ O	CO ₂
Composition	70.15%	28%	1.75%	0.1%

Table 3.3: Composition of Atmosphere
Source: Muhir Kapoor

Air Filter configuration:

- 1. Permanent Magnet Filter:** Filters toxic Martian regolith
- 2. Stainless steel pre filter:** filtration of large particles and debris like dust, hair, etc.
- 3. WO₃ photocatalytic filter with LEDs:** Filtration of viruses, bacteria, molds, VOCs, odors,
- 4. HEPA-13:** Filtration of smoke, dust, pollen, microbes and particles up to 0.3 μm in size.
- 5. TiO₂ photocatalytic filter with UVC:** Filtration of VOCs, ethylene released by plants and microbes up to 0.001 μm in size.

Food production via **Aeroponics** and **cultured meat production**.

Food production will be carried out in greenhouses that trap heat.

Initially, fertilizers would be sourced from resource rich **dark basaltic sands** present in EMC. The extracted fertilizers would be complemented by the nutrients sourced from waste recycling systems

Citrus fruits, vegetables, grains and meats rich in nutrients, such as **vitamins A & C, pectin fiber and omega-3 fatty acids** will be grown locally as they are beneficial countermeasures to damage from long-term radiation exposure. [14]

FIRST HABITAT FOR 4-8 PEOPLE

Human Activities & Plant Growth Chamber

- Inputs: Oxygen, water, nutrients
- Outputs: Carbon dioxide, organic waste, transpiration water vapour

Photobioreactor with *Spirulina platensis*

- Inputs: Carbon dioxide, light (LEDs/natural)
- Processes: Photosynthesis
- Outputs: Oxygen, organic biomass

Anaerobic Digester

- Inputs: Organic waste (human waste, plant biomass)
- Processes: Thermophilic bacterial digestion
- Outputs: Volatile fatty acids, ammonium (NH_4^+), essential minerals

Nitrification Chamber

- Inputs: Ammonium (NH_4^+)
- Processes: Conversion to nitrates by nitrifying bacteria
- Outputs: Nutrients (nitrates)

Atmosphere Management Module

- Inputs: Oxygen, carbon dioxide
- Processes: Circulation and balance maintenance
- Outputs: Optimized air conditions for humans and plants

Atmospheric Condensers

- Inputs: Exhalation, sweat, plant transpiration
- Processes: Condensation of water vapor
- Outputs: Recovered water

Plant Growth Chamber

- Inputs: Water, nutrients, carbon dioxide
- Processes: Crop cultivation, transpiration-driven water recovery
- Outputs: Oxygen, food, water vapor

Water Purification Unit

- Inputs: Wastewater, organic byproducts
- Processes:
 - Microbial desalination
 - Urease bacteria for purification
 - Solid waste drying and microbial degradation
- Outputs: Potable water, additional nutrients

FIRST HABITAT FOR 4-8 PEOPLE

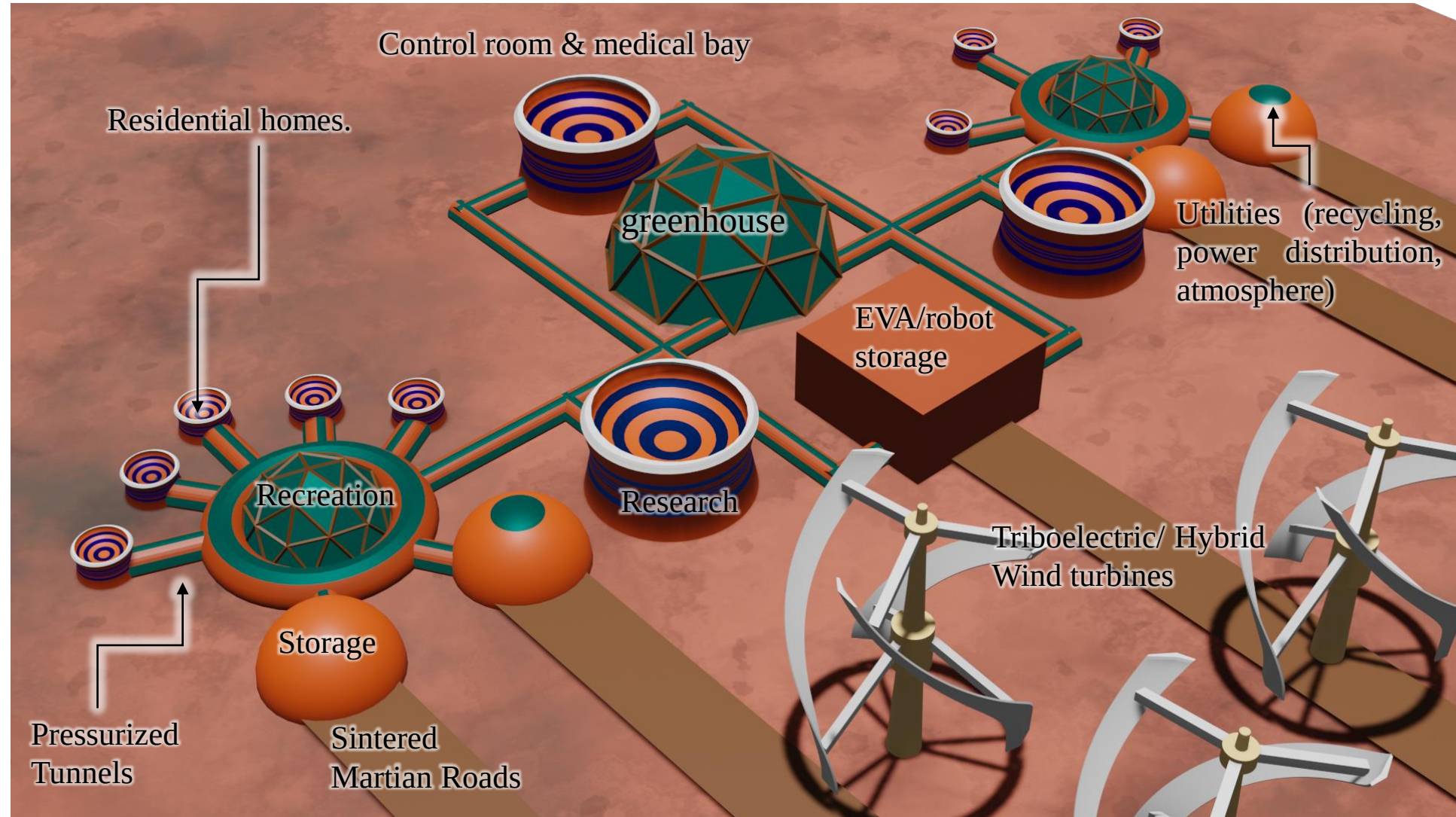


Figure 2.6 Base Layout at IOC
Source: Muhir Kapoor

Phase 1: Habitat Expansion and Essential Systems (2040–2045)

Goal: Accommodate 16 residents with enhanced resource systems and initial self-sufficiency.

Housing:

- Residential modules expanded using basalt fiber composites and cast basalt.
- Airtight pathways and modular layouts ensure connectivity and future adaptability.

Energy Systems:

- Hybrid photovoltaic and wind systems generate 580 kW power.
- Triboelectric turbines installed along canyon walls leverage high wind speeds.
- Compressed energy storage manages fluctuations caused by dust storms.

Food Systems:

- Expanded aeroponic greenhouses meet dietary needs.
- Cultured meat production provides diverse protein sources.

Water and Atmosphere:

- Water vapor adsorption reactors extract water from atmospheric fogs.
- Closed-loop bioregenerative life-support system includes Spirulina photobioreactors for oxygen and nutrient recovery.
- **A pipeline system from Candor Chasma to eastern Melas Chasma begins construction in Phase 1 and is completed in the final phase, ensuring a high-volume water supply for industrial and residential use.**

Scientific Research:

- Remote stations study biosignatures and Martian geology.
- Swarm robotics and ground-penetrating radar map subsurface areas.

Phase 2 (2045–2050)

Goal: Achieve resource independence with expanded infrastructure.

Resource Extraction:

- Swarm robotics mine basaltic sands, hydrated sulfates, and kieserites.
- Biomining recovers metals and nutrients.

Material Processing:

- Refined Martian ores (iron, magnesium, aluminum, silicon) support construction and tools.
- Composite and ceramic materials produced for housing and tools.

Energy Systems:

- Geothermal energy exploration begins in Phase 2, with systems established in high thermal gradient zones by the end of the phase.

Food Systems:

- Microbial milk production systems provide dairy alternatives.

Manufacturing:

- 3D printing uses regolith for tools and structural components.
- Sodium processed for solar coatings; glass and ceramics manufactured for windows and vessels

Phase 3 (2050–2055)

Goal: Expand to 35 residents with improved infrastructure and livability.

Housing:

- Multi-level habitat modules with communal and recreational spaces.
- Treated Martian soil enables **edible landscaping** for mental well-being.

Transportation:

- Automated systems for material transport and **recreational vehicles** for surface exploration.

Medical Facilities:

- **Modular bioreactors cultivate microbes for antibiotics and therapeutic proteins.**

Energy and Utilities:

- **Integrated solar, wind, and geothermal systems generate 1735 kW.**
- Dust mitigation upgraded with self-cleaning materials.

Resource Processing:

- **Biomining of rare earth elements supports advanced electronics.**
- **Flexible materials manufactured from Martian polymers.**

Phase 4 (2055–2060)

Goal: Establish a sustainable village for 50 residents.

Urban Planning:

- Optimized zones for residential, industrial, agricultural, and recreational activities.
- Underground tunnels for contingencies.

Energy Systems:

- Fusion reactors** supplement renewables, generating 3775 kW total.
- Gene-edited microbes produce hydrocarbons for biofuels.

Food and Water Systems:

- Expansive greenhouses grow diverse crops.
- Completed pipelines to candor chaos ensure a high-volume water supply.**

Scientific and Economic Development:

- Advanced labs explore Martian geology and interplanetary technologies.
- Export industries produce propellants and advanced materials.

Manufacturing:

- Locally sourced materials support spacecraft and robotics production.

Unique Features:

- Spacecraft manufacturing and mass driver enable efficient missions.
- Community governance allows resident participation in decision-making

1. Energy Production and Storage

- Self-Sufficient:** Solar and wind systems (locally sourced materials); solid-state batteries and compressed energy storage.
- Earth Dependency:** Initial components for energy systems.
- Dependency Reduction:** ISRU scaling (local production) by Phase 2 (2045–2050); geothermal and fusion reactors by Phase 4 (2055–2060).

2. Water Supply

- Self-Sufficient:** Water from hydrated minerals, atmospheric fog (Phase 1); closed-loop recycling.
- Earth Dependency:** None after initial setup.
- Dependency Reduction:** Subsurface ice extraction (RodWell) by Phase 3 (2050–2055).

3. Food Production

- Self-Sufficient:** Aeroponic greenhouses, cultured meat, basalt-based fertilizers (Phase 1).
- Earth Dependency:** Seeds, cultures, and initial nutrients.
- Dependency Reduction:** Greenhouse expansion (Phase 2); microbial dairy alternatives and genetically engineered crops in later phases.

4. Construction and Manufacturing

- Self-Sufficient:** Local materials (basalt fibers, regolith, metals) and advanced 3D printing (Phase 2).
- Earth Dependency:** High-precision equipment, initial tools.
- Dependency Reduction:** Biomining and rare earth extraction (Phase 3); local ceramics, polymers, and textiles production.

5. Life Support Systems

- Self-Sufficient:** Oxygen via microwave plasma technology; closed-loop bioregenerative systems using cyanobacteria (Phase 1).
- Earth Dependency:** Minimal for initial setup.

Strategies for Reducing Dependency

- ISRU Scaling:** Mining and processing Martian resources for energy, construction, and manufacturing.
 - Automation and Robotics:** AI-driven systems for extraction, construction, and maintenance.
 - Local Innovation:** Adapting Earth technologies and improving ISRU methods.
 - Sustainable Expansion:** Modular designs for energy, water, and food systems.
 - Community Development:** Recreational spaces, biodomes, and Earth-like environments to ensure mental health and settlement viability.
- By 2060:** Self-sustaining village with 50 residents, minimal Earth dependency, and a foundation for interplanetary exploration.

CITATIONS

- [1] [1007.PDF](#)
- [2] [e-melas-ez.pdf](#)
- [3] [Leverage the Power of Swarming Robotics to help NASA Locate Resources, Excavate, and Build on the Moon.](#)
- [4] [ntrs.nasa.gov/api/citations/20190002496/downloads/20190002496.pdf](#)
- [5] [NASA's Mars Chopper Concept \(Rendering\) | NASA Jet Propulsion Laboratory \(JPL\)](#)
- [6] [Development of a Lizard-Inspired Robot for Mars Surface Exploration – PMC](#)
- [7] [In-Situ Resource Utilization \(ISRU\) Living off the Land](#)
- [8] [PowerPoint Presentation](#)
- [9] [ESA - New plasma tech for oxygen, fuel and fertiliser from Mars' atmosphere](#)
- [10] [Fueling a Human Mission to Mars – NASA](#)
- [11] *Martian buildings: structural forms using in-place sources.*
- [12] [A Hybrid Power System for a Permanent Colony on Mars | Space: Science & Technology](#)
- [13] [esmats.eu/esmatpapers/pastpapers/pdfs/2023/pozo.pdf](#)
- [14] [284275main Radiation HS Mod3.pdf](#)